OneEmbedding

A universal codec for PLM per-residue embeddings

Validated across 5 PLMs · 4 task families · with rigorous BCa confidence intervals

Ivan · Rost lab seminar · 2026

The problem

Protein language models emit **per-residue tensors** `(L, D)` per protein:

PLM	D	size for 5K proteins (fp16)
ProtT5-XL	1024	~5 GB
ESM2-650M	1280	~6 GB
ESM-C 600M	1152	~5 GB
ANKH-large	1536	~7 GB

UniProt is **250 M proteins**. Storing, shipping, and searching raw embeddings at scale is impractical.

Goal: compress without losing downstream task performance.

Method — one class, four knobs

```
codec = OneEmbeddingCodec()  # default: 896d binary, ~37x, no codebook
codec.fit(training_embeddings)
data = codec.encode(raw_embeddings)
data['per_residue']   # (L, 896) for per-residue tasks
data['protein_vec']   # (3584,) for retrieval / clustering
```

Pipeline: `center → ABTT (off) → RP to d_out → quantize → DCT K=4 protein vec`

Knob	Default	Why
`d_out`	896	PQ-divisible, Exp 47 direct test
`quantization`	`'binary'`	Exp 47: 94.9% disorder vs PQ128 91.4%, no codebook
`pq_m`	auto (=224 at 896)	~4d sub-vectors
`abtt_k`	0	Exp 45: PC1 73% aligned with disorder

Statistics — Rost-lab convention end-to-end

- **Bootstrap:** BCa (DiCiccio & Efron 1996), B=10 000, percentile fallback for n<25
- **Disorder:** pooled residue-level Spearman ρ (SETH/CAID), cluster bootstrap (resample proteins)
- **Retention: paired** bootstrap — same protein-id resample drives raw and compressed
- **Probes:** CV-tuned (`GridSearchCV` on C/α grids), `random_state=42`
- **Multi-seed:** predictions averaged across {42, 123, 456} **before** bootstrap (Bouthillier 2021)
- **Baselines:** retrieval uses identical DCT K=4 pooling on raw and compressed

Same `metrics.statistics` module backs Exp 43 / 44 / 46 / 47 — no shadow implementations.

Audit-verified: every cited Exp 43/44/46/47 cell is bit-perfect against source JSONs (`docs/_audit/claims.md`).

Headline — Exp 47, ProtT5-XL

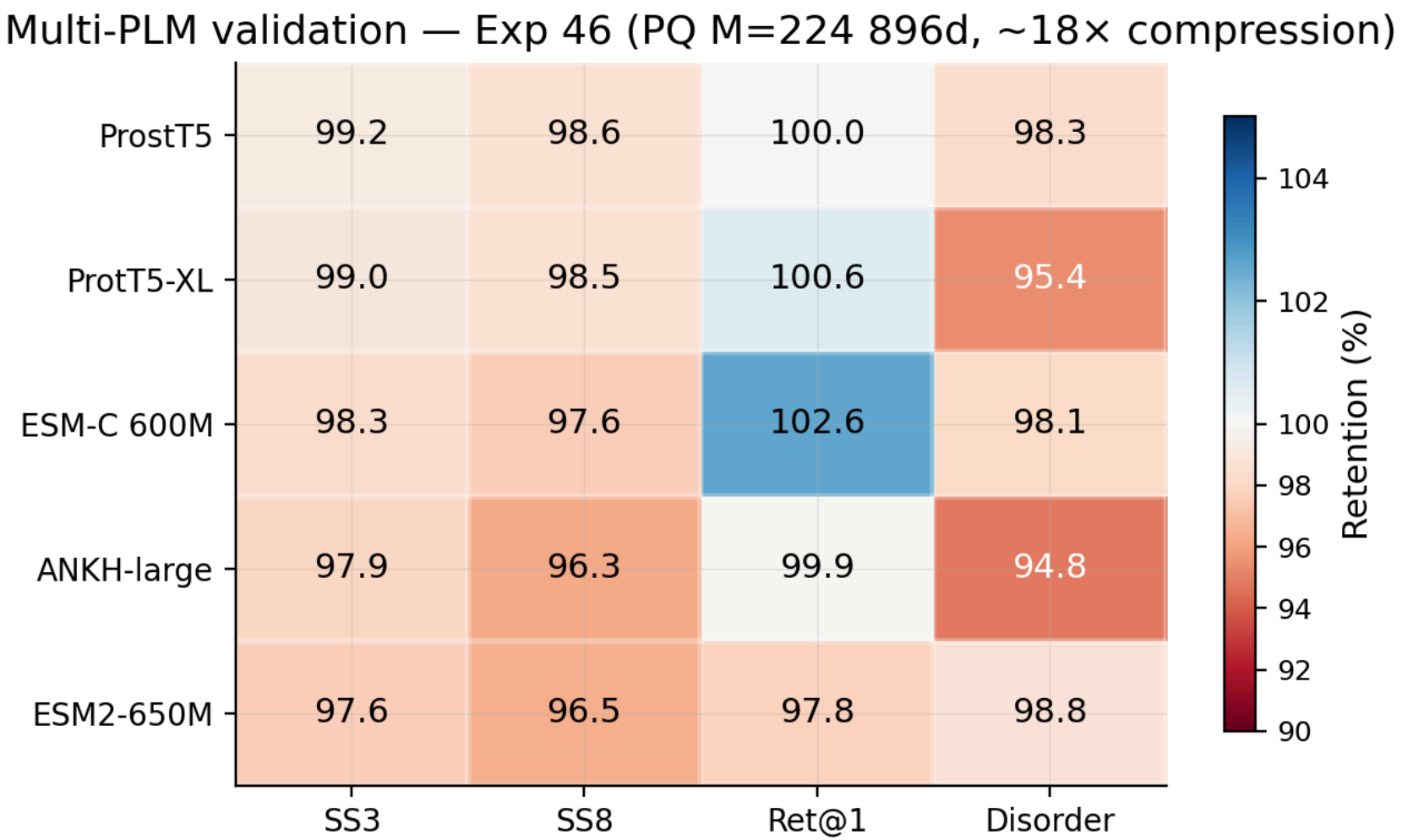
Config	Compression	SS3	SS8	Ret@1	Disorder
lossless 1024d	2×	100.2 %	100.0 %	100.4 %	100.0 %
fp16 896d	2.3×	100.0 %	99.2 %	100.6 %	98.6 %
int4 896d	9×	99.8 %	98.8 %	100.4 %	98.2 %
PQ M=224 896d *(max quality)*	**18×**	**99.0 %**	**98.5 %**	**100.6 %**	**95.4 %**
PQ M=128 896d	32×	97.5 %	96.1 %	100.1 %	91.4 %
binary 896d *(default)*	**37×**	**97.6 %**	**95.0 %**	**100.4 %**	**94.9 %**

Retrieval ≥99.9 % across the entire range. Disorder is the most sensitive cell (4–5 pp gap at 37×).

Pareto

Codec sweep — Exp 47, ProtT5-XL on 4-task mean retention
Mean retention across 4 task families (%)
102
101
100
99
98
97
96
95
94
raw baseline (100 %)
lossless 1024d
fp16 896d
int4 896d
PQ M=224 896d
binary 896d
PQ M=128 896d
10^1
Compression factor (×, log scale)

Multi-PLM validation — Exp 46 (PQ M=224, ~18×)



Same train/test partition per PLM (single split file, embeddings re-extracted). Disorder is consistently the

Receiver-side decode — `numpy + h5py` only

```
import h5py, numpy as np

with h5py.File("protein.one.h5") as f:
    bits   = f["per_residue_bits"][:]      # uint8 packed
    means  = f.attrs["means"][:]
    scales = f.attrs["scales"][:]
    L, D   = int(f.attrs["seq_len"]), int(f.attrs["d_out"])

unpacked = np.unpackbits(bits, axis=1, bitorder="big")[:, :D]
signs    = unpacked.astype(np.float32) * 2 - 1     # {0,1} → {-1,+1}
per_res  = signs * scales + means                   # (L, D) reconstructed
```

No codebook for binary, int4, fp16. PQ correctly requires the codebook.

This is the load-bearing "universal codec" claim. It survives the audit.

What didn't work — and why we kept it

Tried	Verdict	Decision
VQ / RVQ (K=16384)	79 % SS3, **58 % disorder** retention	dropped; not a bug, real ceiling
ABTT (top-PC removal) on by default	Exp 45: PC1 73 % aligned with disorder direction; costs 6–11 pp on disorder	**default** `abtt_k=0`
PCA projection vs RP	Exp 38: RP wins on per-residue retention	random projection chosen
Wavelet, NMF, OT, TDA, SimHash, AA-residual, tensor-train, channel-prune, zstd	None beat the unified codec on Pareto	preserved as archived branches (audit trail)

Three exploratory branches (`worktree-agent-*`) kept in place as audit history — `docs/_audit/worktrees.md` .

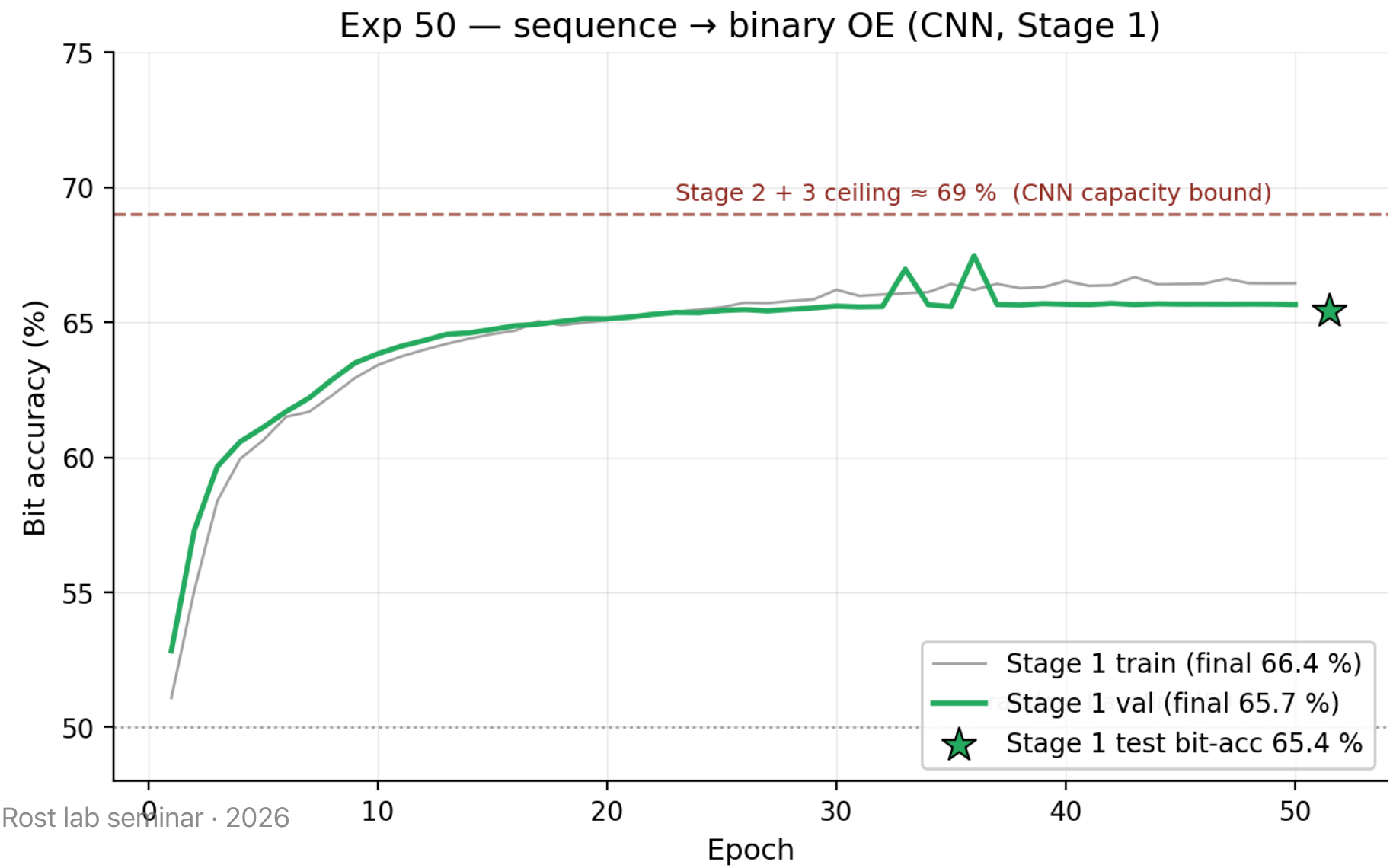
Sequence → binary OE (Exp 50) — setup

Can we predict the compressed (L, 896) binary One Embedding **directly from amino acid sequence** — no PLM at inference time?

- **Architecture:** small dilated CNN (~4.2 M params)
- **Targets:** binary OE bits from `OneEmbeddingCodec(d_out=896, quantization='binary')`
- **Data (sighting):** SCOPe-40 2493 proteins, random 80 / 10 / 10 split
- **Loss:** binary cross-entropy

Honest caveat: random splits leak homology. The rigorous CATH-cluster re-run (H-split + T-split + MMseqs2 audit + 3-seed) is **designed but not yet executed**.

Sequence → binary OE — the ceiling



What's not yet solved

1. **Disorder retention plateaus at ~95 %** (single-PLM rigorous; 94.8–98.8 % across all 5 PLMs)
   → mechanism: Exp 45 ABTT-PC1 finding (geometric, not noise)
   → fix: **Exp 51 PolarQuant** (magnitude-augmented binary, expected +2–3 pp at 36×, no codebook)
2. **Sequence → OE capacity-bound at ~69 %** (Exp 50 ceiling)
   → fix: **Stage 4 transformer** (designed; 3–5 days on Mac)
3. **No co-distilled VESM baseline** — Bromberg lab 2026, MIT-licensed; earmarked
4. **No VEP / ProteinGym evaluation** — classical PLM-quality benchmark; earmarked

Roadmap

Next 2 weeks:

- **Exp 50 Stage 4** — transformer backbone, lift the 69 % ceiling
- **Exp 51 PolarQuant** — close the disorder gap

Next month:

- **Exp 52** — 3Di multi-task head (sequence → binary OE + 3Di tokens)
- **Exp 53** — Foldseek-mined PDB / AFDB training data (only after Stage 4 — Stage 3 already showed 2× data didn't help the CNN)
- **Multi-teacher distillation** — earmarked, revisit after 51-53

Open earmarks: VESM benchmark · VEP / ProteinGym · ProtSpace ESM-vs-ProtT5 cluster comparison

Summary

One Python class, four knobs. Center + RP896 + binary, ~37×, ~17 KB / protein, no codebook.

95–100 % retention across 4 task families × 9 datasets × **5 PLMs**, with rigorous BCa CIs.

Honest weak spots: disorder ~5 pp gap (geometric, not noise), Exp 50 capacity-bound, no VESM baseline yet.

Asks: feedback on PolarQuant design · Stage 4 transformer choice · VEP integration

Pointers: `docs/STATE_OF_THE_PROJECT.md` · `docs/MANUSCRIPT_SKELETON.md` · `docs/EXPECTED_QA.md` · `docs/AUDIT_FINDINGS.md`